

Comparative Analysis of XGBoost and SVM for Identifying Network Flooding Attacks

Academic Machine Learning & Cybersecurity Project

Project Domain	Machine Learning / Cybersecurity
Project Type	Academic Project
Primary Models	XGBoost and Support Vector Machine (SVM)
Additional Models	Decision Tree, Random Forest and LSTM
Focus	Classification and identification of network flooding / DDoS attacks

1. Project Overview

Network flooding and Distributed Denial-of-Service (DDoS) attacks can generate abnormal traffic patterns that disrupt network availability. This project studies machine-learning classification approaches for identifying such traffic and compares multiple models, with particular emphasis on XGBoost and Support Vector Machine (SVM).

2. Problem Statement

The objective is to determine whether machine-learning models can effectively distinguish network traffic associated with flooding attacks and to compare their classification performance using accuracy and statistical analysis.

3. Objectives

- Prepare and split the relevant network-traffic dataset for model development and evaluation.
- Train classification models for identifying network flooding / DDoS traffic.
- Compare XGBoost and SVM performance.
- Extend the comparison with Decision Tree, Random Forest and LSTM models.
- Evaluate model performance using accuracy and statistical analysis.

This concise report is prepared from the academic project poster and is intended as a portfolio/repository artifact.

Methodology and Implementation

4. Methodology

Data Collection & Preparation	Collect and prepare network-traffic data relevant to flooding/DDoS detection.
Data Splitting	Separate the available data into appropriate portions for model training and evaluation.
Model Training	Train the selected machine-learning classification models.
Prediction	Use trained models to classify network traffic.
Performance Evaluation	Compare model performance using accuracy and statistical analysis.

5. Models Considered

XGBoost: Primary comparison model and gradient-boosting classifier.

Support Vector Machine (SVM): Classical supervised classifier compared against XGBoost.

Decision Tree: Additional baseline/comparison classifier.

Random Forest: Ensemble tree-based comparison model.

LSTM: Sequence-oriented deep-learning comparison model.

6. Technical Approach

- The project applies supervised machine-learning classification to a practical cybersecurity problem.
- Model outputs are evaluated on evaluation/test data rather than relying only on training performance.
- The study compares several models instead of evaluating a single algorithm in isolation.
- XGBoost is used as the main reference model across the comparisons shown in the project poster.

7. Project Contribution

The project demonstrates the application of machine learning to network-security classification and provides a comparative view of classical, ensemble and sequence-based approaches.

Results, Discussion and Conclusion

8. Reported Results

Model	Reported mean accuracy
XGBoost	97.1%
SVM	96.8%
Decision Tree	82.3%
Random Forest	86.9%
LSTM	85.9%

The values above are the mean accuracy figures reported in the supplied academic poster; they are presented as project-reported results rather than independently reproduced benchmarks.

9. Discussion

- XGBoost achieved the highest reported mean accuracy among the compared models.
- The XGBoost and SVM comparison showed a smaller performance gap than the comparisons involving the other models.
- The results suggest that boosting-based classification can be effective for this network-attack classification task.
- Model selection should also consider precision, recall, false positives, computational cost and generalization, not accuracy alone.

10. Conclusion

The project evaluated multiple machine-learning approaches for identifying network flooding attacks and found XGBoost to be the strongest-performing model among the approaches reported in the project poster. The work demonstrates how supervised machine learning can support network-security monitoring.

11. Future Scope

- Evaluate additional attack classes and more diverse network environments.
- Include precision, recall, F1-score and confusion matrices alongside accuracy.
- Tune model hyperparameters and examine feature importance.
- Test models on larger and more recent datasets.
- Explore real-time deployment for network monitoring.

Repository: [network-flooding-attack-ml-analysis](#)